

DISCRETE MATHEMATICS

EXERCISES — GENERATING FUNCTIONS

(SUPPLEMENTARY)

Exercise 1

Find the ordinary generating function $f(x) = \sum_{n=0}^{\infty} a_n x^n$ for each sequence.

- (a) The Fibonacci sequence $F_0 = 0$, $F_1 = 1$, $F_n = F_{n-1} + F_{n-2}$ for $n \geq 2$. (*Hint.* Write the GF equation directly from the recurrence.)
- (b) $a_n = n\alpha^n$ for fixed $\alpha \in \mathbb{R}$, $n = 0, 1, 2, \dots$ (*Hint.* Differentiate the geometric series.)
- (c) The triangular numbers: $a_n = \frac{n(n+1)}{2}$, $n = 0, 1, 2, \dots$
- (d) $a_n = \sum_{k=0}^n F_k$ (partial sums of Fibonacci numbers), where F_k are as in (a). Express the answer in terms of the GF found in (a).

Exercise 2

Suppose $f(x) = \sum_{n=0}^{\infty} a_n x^n$. Express the generating function $F(x)$ for each derived sequence in terms of f , f' , f'' .

- (a) $A_n = n(n-1)a_n$
- (b) $A_n = a_n + a_{n+1}$ (*Hint.* Use the shift formula from the standard manipulation exercises.)
- (c) $A_n = \sum_{k=0}^n (n-k)a_k$ (*Hint.* This is the convolution of $\{n\}$ with $\{a_n\}$.)
- (d) $A_n = a_n - 2a_{n-1} + a_{n-2}$ for $n \geq 2$, $A_0 = a_0$, $A_1 = a_1 - 2a_0$.

Exercise 3

Find the indicated coefficient.

- (a) The coefficient of x^{15} in $(1 + x + x^2 + x^3 + x^4 + x^5)^6$.
- (b) The coefficient of x^{15} in $\frac{x}{1-x-x^2}$. (*Hint.* Recall Exercise 1(a).)
- (c) The coefficient of x^{15} in $\frac{x^4}{(1-x)^3}$.
- (d) The coefficient of x^n in $\frac{1}{(1-x)^2(1-x^2)}$ for general n . Simplify separately for even and odd n .

Exercise 4

Third-order linear recurrence. Use generating functions to find a closed-form expression for a_n given

$$a_n = 3a_{n-1} - 3a_{n-2} + a_{n-3}, \quad n \geq 3, \quad a_0 = 0, \quad a_1 = 1, \quad a_2 = 4.$$

(*Hint.* The characteristic polynomial factors as a perfect cube. The resulting GF can be written as $\frac{x+x^2}{(1-x)^3}$; use partial fractions or a known series to extract $[x^n]$.)

Exercise 5

System of recurrences. Two sequences $\{a_n\}$ and $\{b_n\}$ satisfy

$$a_n = 2a_{n-1} + b_{n-1}, \quad b_n = a_{n-1} + 2b_{n-1}, \quad n \geq 1,$$

with initial conditions $a_0 = 1$ and $b_0 = 0$.

- (a) Let $F(x) = \sum a_n x^n$ and $G(x) = \sum b_n x^n$. Write the system of linear equations relating F and G .
- (b) Eliminate G to obtain a single equation for F , and solve for $F(x)$.
- (c) Find a closed form for a_n . Then find $G(x)$ and a closed form for b_n .
- (d) Verify your answers for $n = 0, 1, 2, 3$.

Exercise 6

Catalan numbers. The Catalan numbers C_n count the number of ways to fully parenthesise a product of $n+1$ factors. They satisfy

$$C_n = \sum_{k=0}^{n-1} C_k C_{n-1-k}, \quad n \geq 1, \quad C_0 = 1.$$

- (a) Show that the generating function $f(x) = \sum_{n=0}^{\infty} C_n x^n$ satisfies the quadratic functional equation

$$f(x) = 1 + x f(x)^2.$$

- (b) Solve for $f(x)$ and show that the appropriate branch is

$$f(x) = \frac{1 - \sqrt{1 - 4x}}{2x}.$$

- (c) Use the formula $C_n = \frac{1}{n+1} \binom{2n}{n}$ to compute C_0, C_1, C_2, C_3, C_4 .
- (d) Verify that $C_3 = 5$ agrees with direct counting: list the five parenthesisations of $a \cdot b \cdot c \cdot d$.

Exercise 7

Tiling a board. Let a_n denote the number of ways to tile a $1 \times n$ board using tiles of size 1×1 and 1×2 (the tiles may not be rotated or overlapped; the board is completely covered).

- (a) Show that $a_n = a_{n-1} + a_{n-2}$ for $n \geq 2$, with $a_0 = 1$ and $a_1 = 1$.
- (b) Use generating functions to find the closed-form GF for $\{a_n\}$.
- (c) Express a_n in terms of Fibonacci numbers.
- (d) Find the number of tilings of a 1×8 board.

Exercise 8

Counting compositions. An *ordered composition* of n into k parts is a k -tuple $(e_1, \dots, e_k)$ of positive integers with $e_1 + \dots + e_k = n$.

- (a) Find the generating function for the number of compositions of n into exactly k positive parts (as a function of k and x).
- (b) Show that the total number of compositions of n (summing over all $k \geq 1$) is 2^{n-1} for $n \geq 1$. (*Hint.* Sum the GFs over k and extract the coefficient of x^n .)
- (c) Find the number of compositions of 10 into parts equal to 1, 2, or 3. (*Hint.* Adapt the tiling GF.)

Exercise 9

Partial fractions and closed forms.

- (a) Find the partial fraction decomposition of $\frac{3x-1}{(1-x)(1-2x)(1-3x)}$, and hence find a closed form for the sequence a_n with generating function equal to this expression.
- (b) The generating function of a sequence $\{a_n\}$ is $f(x) = \frac{x^2}{(1-x)^2(1+x)}$. Find a_n in closed form. (*Hint.* Factor the denominator and use partial fractions with the cases $\frac{A}{1+x}$, $\frac{B}{1-x}$, $\frac{C}{(1-x)^2}$.)

Exercise 10

Identities via generating functions. Prove the following combinatorial identities using generating functions.

- (a) $\sum_{k=0}^n k \binom{n}{k} = n \cdot 2^{n-1}$.
(*Hint.* Differentiate the identity $(1+x)^n = \sum_k \binom{n}{k} x^k$.)

(b) $\sum_{k=0}^n \binom{n}{k} \binom{n}{n-k} = \binom{2n}{n}.$

(*Hint.* This is the Vandermonde identity; examine $[x^n](1+x)^{2n}.$)

(c) $F_1 + F_2 + \cdots + F_n = F_{n+2} - 1$, where F_n is the n -th Fibonacci number.

(*Hint.* Use the GF from Exercise 1(a) and the partial-sum formula from Exercise 1(d).)